

Matrices & Vectors

ANSWER KEY



Addition and scalar multiplication

- 1 Let A be the matrix with rows $\langle 2, 1 \rangle$ and $\langle 3, -1 \rangle$, and B be the matrix with rows $\langle -1, 4 \rangle$ and $\langle 0, 2 \rangle$. Find $A + B$ and $3A$.

$A + B$ has rows $\langle 1, 5 \rangle$ and $\langle 3, 1 \rangle$. $3A$ has rows $\langle 6, 3 \rangle$ and $\langle 9, -3 \rangle$.

- 2 Using the same A and B , find $A - 2B$.

$A - 2B$ has rows $\langle 2 - (-2), 1 - 8 \rangle = \langle 4, -7 \rangle$ and $\langle 3 - 0, -1 - 4 \rangle = \langle 3, -5 \rangle$.

Matrix multiplication

- 3 Using A (rows $\langle 2, 1 \rangle, \langle 3, -1 \rangle$) and B (rows $\langle -1, 4 \rangle, \langle 0, 2 \rangle$) from above, find AB .

AB has rows $\langle -2, 10 \rangle$ and $\langle -3, 10 \rangle$: e.g. row 1, column 1 is $(2)(-1) + (1)(0) = -2$.

- 4 Let A have rows $\langle 2, 1 \rangle$ and $\langle 3, -1 \rangle$, and let $\mathbf{v} = \langle 2, -3 \rangle$. Find $A\mathbf{v}$.

$A\mathbf{v} = \langle (2)(2) + (1)(-3), (3)(2) + (-1)(-3) \rangle = \langle 1, 9 \rangle$.

- 5 Explain, without computing BA , why AB and BA need not be equal for 2×2 matrices.

Matrix multiplication combines rows of the first factor with columns of the second; swapping the order pairs different rows with different columns, so in general $AB \neq BA$ (matrix multiplication is not commutative).

Determinant and inverse

- 6 Find $\det A$ for the matrix A with rows $\langle 2, 1 \rangle$ and $\langle 3, -1 \rangle$, and state whether A is invertible.

$\det A = (2)(-1) - (1)(3) = -5 \neq 0$, so A is invertible.

- 7 Find A^{-1} for the matrix A with rows $\langle 2, 1 \rangle$ and $\langle 3, -1 \rangle$.

$A^{-1} = \frac{1}{-5} \times$ the matrix with rows $\langle -1, -1 \rangle$ and $\langle -3, 2 \rangle$, i.e., A^{-1} has rows $\langle 0.2, 0.2 \rangle$ and $\langle 0.6, -0.4 \rangle$.

- 8 Without computing it, explain why the matrix with rows $\langle 4, 2 \rangle$ and $\langle 6, 3 \rangle$ has no inverse.

$\det = (4)(3) - (2)(6) = 12 - 12 = 0$, and a matrix with determinant 0 has no inverse (Theorem 2.1).

Solving systems with the inverse matrix

- 9 Solve the system $3x - y = 8$ and $x + 2y = 5$ using $\mathbf{x} = M^{-1}\mathbf{b}$.

M has rows $\langle 3, -1 \rangle$ and $\langle 1, 2 \rangle$, so $\det M = (3)(2) - (-1)(1) = 7$, and M^{-1} has rows $\langle \frac{2}{7}, \frac{1}{7} \rangle$ and $\langle -\frac{1}{7}, \frac{3}{7} \rangle$. Then $x = \frac{1}{7}[2(8) + 1(5)] = 3$ and $y = \frac{1}{7}[-1(8) + 3(5)] = 1$. Check: $3(3) - 1 = 8$ and $3 + 2(1) = 5$.

Vector magnitude and dot product

- 10 For $\mathbf{u} = \langle 3, 4 \rangle$ and $\mathbf{w} = \langle -4, 3 \rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.

$$\mathbf{u} \cdot \mathbf{w} = (3)(-4) + (4)(3) = 0, \text{ and } |\mathbf{u}| = \sqrt{3^2 + 4^2} = 5.$$

- 11 What does $\mathbf{u} \cdot \mathbf{w} = 0$ tell you about $\mathbf{u}$ and $\mathbf{w}$ from the previous question?

A zero dot product means $\mathbf{u}$ and $\mathbf{w}$ are perpendicular (orthogonal).

Angle between vectors

- 12 Find the angle between $\mathbf{p} = \langle 1, 0 \rangle$ and $\mathbf{q} = \langle 1, 1 \rangle$.

$$\mathbf{p} \cdot \mathbf{q} = 1, |\mathbf{p}| = 1, |\mathbf{q}| = \sqrt{2}. \cos \theta = \frac{1}{\sqrt{2}} \approx 0.7071 \Rightarrow \theta = 45^\circ.$$

- 13 Find the angle between $\langle 6, -8 \rangle$ and $\langle 3, 4 \rangle$, rounded to the nearest tenth of a degree.

$$\text{Dot product} = 18 - 32 = -14, \text{ magnitudes } 10 \text{ and } 5. \cos \theta = \frac{-14}{50} = -0.28 \Rightarrow \theta \approx 106.3^\circ.$$